

Easy-to-Hard German Continued Pretraining for a Small English Masked Language Model

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Abstract

This proposal studies whether a small English masked language model can adapt to German more effectively when German training data are presented in an easy-to-hard curriculum rather than as a mixed-difficulty stream. The original project idea involved linguistically targeted masking, such as POS-specific masking, but this would require a comparatively large preprocessing and annotation pipeline. I therefore narrow the scope to a more feasible variable: the ordering of German input by readability level. The experiment uses German4All, a readability-controlled German paraphrasing dataset, and continued pretraining of `prajjwall/bert-tiny` with an English BERT tokenizer in Google Colab. A first MVP has already run. It successfully compared a mixed baseline with a strict easy-to-hard curriculum, but the result did not support a clear curriculum advantage: the mixed baseline achieved lower final validation loss overall, while the curriculum condition showed only phase-specific improvements and a very small advantage on the hard split. The next iteration will improve experimental control by saving all outputs to Google Drive, adding step-0 evaluation, using full final validation, balancing difficulty exposure, and testing softer replay-based curricula.

1 Introduction

Masked language modeling (MLM) is usually trained on randomly sampled text, but human second-language learning is rarely organized as random exposure. Learners typically encounter simpler input before more complex input. This project asks whether a similar easy-to-hard principle can help a small English encoder model adapt to German through continued MLM pretraining.

The original project scope considered several linguistically motivated masking strategies, including POS-specific masking and grammar-oriented masking. However, these variants would require reliable POS tagging and additional preprocessing, making the project too large for a small course project. I therefore reduce the scope to a simpler but still meaningful comparison: readability-based curriculum training versus mixed-difficulty training.

The goal is not to train a competitive German model. Instead, the project tests whether data order has a measurable effect under a fixed compute budget. This makes the project suitable for Google Colab and for a 3 ECTS term project. The central contribution is a controlled small-scale empirical study of whether a pedagogically plausible curriculum helps cross-lingual continued pretraining.

2 Research Question and Hypotheses

The main research question is:

Does easy-to-hard continued pretraining help a small English masked language model adapt to German more effectively than mixed-difficulty continued pretraining?

I define three working hypotheses.

H1: final-loss curriculum advantage. Under the same training budget, an easy-to-hard curriculum will produce lower MLM validation loss than mixed-difficulty training, especially on medium and hard German validation data.

H2: phase-specific curriculum effect. Even if the final loss is not better, the curriculum condition may show stronger local improvement on the validation split corresponding to the current training phase. For example, during the easy phase, easy validation loss may decrease faster.

H3: mixed exposure as a strong alternative. Mixed training may perform similarly or better because it exposes the model to the full German difficulty distribution from the beginning. This hypothesis became more important after the first MVP result.

3 Data

The experiment uses German4All, a German readability-controlled paraphrasing dataset. It is suitable for this project because it provides German text across five readability levels and aligned paraphrases, which helps reduce topic differences across difficulty levels. These properties make it possible to construct easy, medium, and hard German input without building a new annotated corpus.

For the MVP, the five German4All readability levels are collapsed into three broad difficulty groups. Table ?? shows the resulting data sizes in the first completed run.

Group	Train examples	Validation examples
Easy	45,828	5,090
Medium	22,914	2,545
Hard	45,828	5,090
Total	114,570	12,725

Table 1: German4All difficulty groups used in MVP v1.0.

The main advantage of German4All is feasibility: it directly supports a curriculum design and has enough data for a small continued-pretraining experiment. Its main limitation is that readability labels are designed for human readers, not necessarily for MLM difficulty. A sentence that is easy for a human German learner may still be difficult for an English WordPiece tokenizer because of German capitalization, compounds, inflection, umlauts, and token fragmentation. For this reason, future work may use purely human-authored CEFR-style data, but the current project keeps German4All as the only dataset to maintain a manageable scope.

4 Model and Training Objective

The MVP uses `prajjwall/bert-tiny` with the `bert-base-uncased` tokenizer. This model is small enough for free Google Colab GPU sessions and allows rapid debugging. The English tokenizer is deliberately kept because the project studies adaptation from an English model to German. However, this also introduces a tokenizer mismatch that must be treated as a limitation.

The training objective is standard masked language modeling. Given an input sequence $x = (x_1, \dots, x_n)$ and a set of masked positions M , the model minimizes

$$\mathcal{L}_{\text{MLM}}(\theta) = -\frac{1}{|M|} \sum_{i \in M} \log p_{\theta}(x_i \mid x_{\setminus M}). \quad (1)$$

Validation loss is measured separately for easy, medium, hard, and overall validation splits.

In MVP v1.0, the training budget was 900 optimization steps, batch size 8, maximum sequence length 128, MLM probability 0.15, and evaluation every 100 steps. Each validation split was evaluated with a maximum of 30 batches during the pilot run.

5 Experimental Design

The first MVP compares two conditions under the same model, tokenizer, masking ratio, sequence length, optimizer, and total number of training steps.

Mixed baseline. The model is trained on easy, medium, and hard examples mixed throughout all 900 steps.

Strict easy-to-hard curriculum. Training is divided into three blocks: steps 1–300 use easy examples, steps 301–600 use medium examples, and steps 601–900 use hard examples.

The main metric is final MLM validation loss. Secondary analysis examines the validation-loss curves by difficulty split in order to detect phase-specific changes.

6 MVP v1.0 Results

The first MVP successfully ran the intended comparison. However, the result does not support H1. Table ?? shows the final validation losses.

Condition	Easy	Medium	Hard	Overall
Curriculum	3.8651	4.1099	4.2101	4.0513
Mixed	3.6923	4.0939	4.2223	3.9983
Curr.-Mixed	+0.1728	+0.0160	-0.0122	+0.0530

Table 2: Final validation loss in MVP v1.0. Lower is better.

The mixed baseline achieved lower final loss on the easy, medium, and overall validation splits. The curriculum model was only slightly better on the hard split, and the difference was very small. Therefore, the current result should be interpreted as a negative or mixed pilot result rather than as evidence for a curriculum advantage.

However, H2 receives partial support. The curriculum schedule produced phase-specific learning behavior: during the easy phase, the model improved faster on the easy validation split; during later phases, it improved more on the corresponding harder splits. This indicates that the curriculum schedule affected training locally, even though these local effects did not accumulate into better final overall performance.

7 Interpretation and Limitations

The most plausible interpretation is that a strict block curriculum is not automatically better for German MLM adaptation. Mixed training is a strong baseline because the model sees the full distribution of German difficulty levels from the beginning. This may allow it to learn basic German patterns while also adapting early to harder lexical and structural patterns.

Several limitations remain. First, the curriculum is too strict: easy examples disappear after phase one, and medium examples disappear after phase two. Human language learning more often involves replay, where easier material remains available while harder material is introduced. Second, difficulty exposure is confounded with order. The mixed baseline follows the natural expanded data ratio of approximately 40% easy, 20% medium, and 40% hard, whereas the curriculum uses equal one-third stage lengths. Third, the experiment uses only one seed, a small model, limited validation batches, and a short 900-step budget. Fourth, human readability may not match MLM difficulty under an English tokenizer.

8 Next Iteration

The next iteration will prioritize experimental control and reproducibility rather than immediately scaling up the model.

Engineering improvements. All logs, checkpoints, configuration files, plots, prepared data, and final tables will be saved to Google Drive by default. This is necessary because Colab local storage can be lost after runtime interruption.

Evaluation improvements. The next notebook will add true step-0 evaluation before any optimizer update. It will also keep fast batch-limited evaluation during training but run full final validation at the final checkpoint. If feasible, fixed masked validation datasets will be used to reduce evaluation noise.

Experimental improvements. I plan four follow-up comparisons:

1. rerun mixed versus strict easy-to-hard with Drive saving and full final validation;
2. compare balanced mixed training against balanced easy-to-hard training to isolate order from exposure ratio;
3. add a hard-to-easy control condition to test whether the direction of the curriculum matters;
4. test a soft replay curriculum, for example 80/15/5 easy-medium-hard in phase 1, 40/40/20 in phase 2, and 20/30/50 in phase 3.

I will also add diagnostics to test whether German4All readability levels correspond to actual MLM difficulty. These diagnostics include initial MLM loss by readability level, mean WordPiece tokens per sentence, subwords per whitespace word, frequent WordPieces by level, and sentence length by level.

9 Expected Contribution

The project can still produce a useful result even if the curriculum hypothesis remains unsupported. A negative result would show that human-readable easy-to-hard ordering does not necessarily improve MLM continued pretraining, at least for a small English model adapting to German with German4All. A stronger positive result would require the next iteration to show that curriculum training remains competitive after controlling for exposure ratio and evaluation noise. In either case, the project contributes a small, reproducible experiment connecting curriculum learning, continued pretraining, simplified-language data, and second-language learning motivation.

10 Conclusion

This project has been narrowed to a feasible comparison between easy-to-hard and mixed German continued pretraining. The first MVP shows that the training pipeline works, but strict block curriculum training did not outperform the mixed baseline. The next step is therefore not to scale immediately, but to make the experiment more reliable: save outputs persistently, evaluate more carefully, control exposure ratio, and test replay-based curricula.

References

- Anschütz, M. et al. 2025. German4All: A Dataset and Model for Readability-Controlled Paraphrasing in German.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of NAACL-HLT*.
- Ronen Eldan and Yuanzhi Li. 2023. TinyStories: How Small Can Language Models Be and Still Speak Coherent English?
- Suchin Gururangan et al. 2020. Don't Stop Pretraining: Adapt Language Models to Domains and Tasks. In *Proceedings of ACL*.
- Minjin Lee et al. 2022. Efficient Pre-training of Masked Language Model via Concept-based Curriculum Masking.